

Mini Project 1.3: Generation Recommender — Test Results

Output of: python recommender.py (8 built-in test cases)

Requirement: "I need low-latency control for a robotic arm"

-> First addressed by: 5G (URLLC) (matched: low-latency, robotic arm)

Why: 5G's URLLC service class targets ~1 ms latency and very high reliability for control-type a

Requirement: "Send text messages and make secure, encrypted calls while roaming abroad"

-> First addressed by: 2G (matched: text messag, encrypt, roaming)

Why: 2G (GSM) digitized voice, added SMS and encryption, and standardized international roaming,

Requirement: "Browse the web and make video calls from my phone"

-> First addressed by: 3G (matched: browse the web, video call)

Why: 3G was designed for mobile internet and multimedia (video calls, web) over packet-switched

Requirement: "Monitor thousands of low power soil sensors on a farm"

-> First addressed by: 5G (mMTC) (matched: sensor, low power)

Why: 5G's mMTC service class supports very dense, low-power, low-rate device deployments.

Requirement: "Holographic telepresence with integrated radio sensing"

-> First addressed by: 6G (vision/research) (matched: holograph, sensing)

Why: 6G is still a research vision: THz spectrum, integrated sensing and communication, and AI-n

Requirement: "Analog cellular voice calls from a moving car"

-> First addressed by: 1G (matched: analog cellular, cellular voice)

Why: 1G introduced cellular frequency reuse and handover for analog voice, though with poor secu

Requirement: "Stream HD video and also perform remote surgery"

-> Does not map to a single generation (2 classes matched):

* 4G (matched: hd video): 4G (LTE) delivered an all-IP, OFDMA-based broadband network that made

* 5G (URLLC) (matched: remote surgery): 5G's URLLC service class targets ~1 ms latency and very

Advice: split the requirement; the latest generation listed is needed to satisfy it all.

Requirement: "A better mobile experience"

-> No clear match. Rephrase with specifics (latency, device count, data rate, voice/data, security)